IMPLEMENTATION OF SYSTOLIC CONVOLUTION ARRAY IN FPGA USING VERILOG

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BONAFIDE CERTIFICATE

This is to certify that the project titled **IMPLEMENTATION OF SYSTOLIC CONVOLUTION ARRAY IN FPGA USING VERILOG** is a bonafide record of work done by

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In partial fulfillment of the requirements for winter internship conducted in the Department of **ELECTRONICS AND COMMUNICATION ENGINEERING** of the **NATIONAL INSTITUTE OF TECHNOLOGY TIRUCHIRAPPALLI,** during the year 2025

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ABSTRACT

A systolic convolution array provides a high-speed and resource-efficient method for performing convolution operations in digital signal processing and deep-learning workloads. In this work, a systolic convolution array is designed and implemented using Verilog HDL on a Xilinx FPGA board. The architecture uses a chain of Processing Elements (PEs) arranged in a pipelined systolic structure, enabling continuous data flow and reduced latency. Input feature maps and kernel values move rhythmically through the array, allowing multiply-accumulate (MAC) operations to occur in parallel. The design is synthesised on a Xilinx FPGA, and parameters such as clock frequency, array size, data width, pipeline depth, and resource consumption are analysed. The effect of PE arrangement, timing optimisation, and memory-mapping strategy is also studied. The objective of this study is to determine the optimum systolic array configuration that provides maximum throughput while minimizing hardware utilization.

Keywords: Systolic array; Convolution; FPGA; Verilog; Parallel processing; MAC units

TABLE OF CONTENTS

BONAFIDE CERTIFICATE **1**
ABSTRACT **2**
INTRODUCTION **4**
Systolic Array–Based Convolution Using EYERISS Row-Stationary Dataflow (3×3 Kernel) **5**
1. Introduction **5**
2. EYERISS Row-Stationary Dataflow: Overview **5**
3. Hardware Architecture **6**
3.1 Processing Element (PE) Architecture **6**
3.2 3×N Systolic PE Array **6**
3.3 Line Buffers (Row Buffers) 7
4. Dataflow Explanation (Cycle-by-Cycle Behavior) **7**
Step 1 — Input Streaming **7**
Step 2 — Weight Preloading **8**
Step 3 — MAC Computation inside the Array **8**
Step 4 — Sliding Window Operation **8**
5. Output Generation **8**
5.1 Some convolution outputs using standard filters **9**
6. Performance Characteristics **10**
7. Pipelining **12**
8. Summary **13**
9. Conclusion **13**
References **15**

INTRODUCTION

Field-Programmable Gate Arrays (FPGAs) have become a preferred platform for accelerating data-intensive computations due to their fine-grained parallelism, hardware reconfigurability, and significantly lower power consumption compared to general-purpose processors. Convolution operations, which dominate the computational load in image-processing and neural-network applications, are especially well-suited for FPGA acceleration because of their repetitive multiply-accumulate (MAC) structure. Traditional RTL-based development allows highly optimized hardware architectures but demands substantial design and verification effort. As a result, identifying efficient FPGA-based convolution architectures that maximise performance while minimizing resource consumption remains an important area of research.

Systolic arrays provide a regular, pipelined computation model in which data moves rhythmically through a network of Processing Elements (PEs). This structured dataflow enables massive parallelism and deterministic timing, making systolic arrays ideal for accelerating convolution kernels. In this project, a systolic convolution array is designed and implemented using Verilog HDL on a Xilinx FPGA board to perform 2D convolution with a 3×3 kernel on a 64×64 grayscale image. The design uses a chain of MAC-based PEs that propagate pixel and kernel values across the array, achieving continuous throughput once the systolic pipeline is filled.

The targeted Xilinx FPGA provides strong hardware capabilities that support systolic acceleration. Typical boards such as the Xilinx Artix-7 or Zynq-7000 series offer:

- 100k–200k LUTs, enabling large parallel architectures
- 240–900 DSP slices for high-speed MAC operations
- Up to 4.8 MB Block RAM (BRAM) for buffering image tiles and weights
- Clock frequencies ranging from 100–200 MHz for stable real-time processing
- Low power consumption (2–5 W) compared to GPU implementations

These hardware features make the FPGA well-suited for implementing systolic convolution, where DSP slices handle continuous MAC operations while BRAM stores intermediate pixel streams.

The objective of this work is to evaluate systolic dataflow as an efficient hardware architecture for image convolution. This includes PE design, data-feeding mechanisms, timing optimisation, and resource-usage analysis. Performance metrics such as latency, throughput, DSP utilisation, and BRAM consumption are measured on the Xilinx board. By demonstrating real-time convolution on a 64×64 image, this project highlights the effectiveness of systolic arrays for accelerating small-kernel image-processing tasks and provides insights for scaling toward larger images, deeper pipelines, and more complex machine-learning workloads.

Systolic Array–Based Convolution Using EYERISS Row-Stationary Dataflow (3×3 Kernel)

1. Introduction

Convolution is the core operation in Convolutional Neural Networks (CNNs) used for image processing and feature extraction. Traditional software implementations perform convolution by sliding a kernel over the input feature map and computing multiply–accumulate (MAC) operations.
However, software convolution suffers from high latency and memory bandwidth limitations when dealing with large feature maps.

To address this, we implemented a hardware systolic-array convolution architecture inspired by the EYERISS accelerator (by MIT) that uses a Row-Stationary (RS) Dataflow. This method optimizes data reuse inside the processing elements (PEs) and minimizes external memory accesses.

Our design performs 3×3 convolution on grayscale images using:

- A systolic PE array (3 rows × N columns depending on image width)
- Local line buffers to stream data
- Weight-stationary and partial-sum stationary datapaths
- Sliding window movement across the image

2. EYERISS Row-Stationary Dataflow: Overview

EYERISS introduced a Row-Stationary Dataflow, which is a mapping strategy that maximizes reuse of

- Input pixels
- Filter weights
- Partial sums

within the PE array.
In a 3×3 convolution, each output pixel is given by:

$$O(x, y) = \sum_{i=0}^{2} \sum_{j=0}^{2} W(i, j) \cdot I(x + i, y + j)$$

The RS method keeps a single row of the 3×3 kernel stationary inside each PE row, while input pixels flow horizontally and partial sums propagate vertically.

- Each input pixel participates in 3 MACs vertically

- Each weight row participates in many horizontal convolutions
- Partial sums require vertical accumulation

The RS dataflow leverages all three reuse types.

3. Hardware Architecture

3.1 Processing Element (PE) Architecture

Each PE contains:

- One weight register (holds one kernel coefficient)
- One input register (streams pixel data)
- One MAC unit
- A partial sum accumulator
- Output forwarding logic (to next PE or next row)

Operations per cycle:

1. Receive input pixel from left
2. Multiply by stationary kernel weight
3. Add to vertical partial sum
4. Pass input pixel to the right
5. Pass partial sum downward

3.2 3×N Systolic PE Array

Because the kernel is 3×3, we use **3 PE rows**, each row holding one row of kernel weights:

Row 0 → W00, W01, W02

Row 1 → W10, W11, W12

Row 2 → W20, W21, W22

Across columns, the array extends to match the width of the input minus 2, so that convolutions stream continuously.

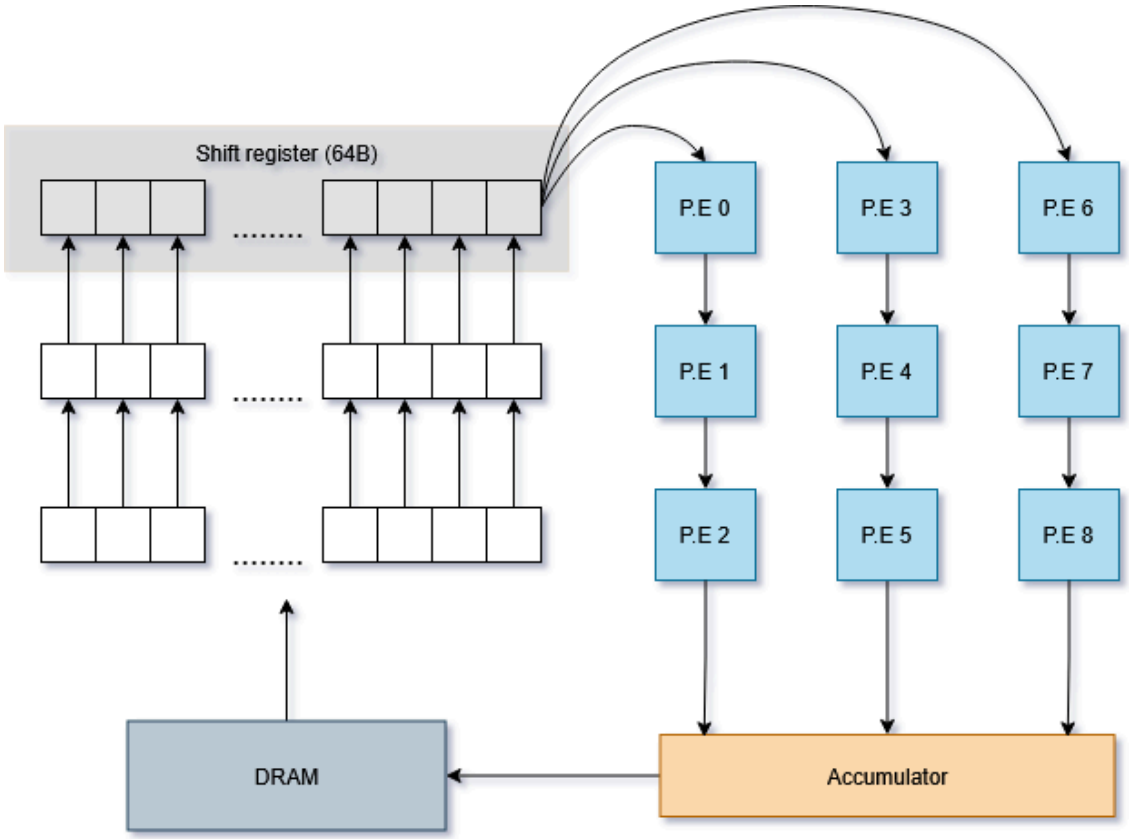


Figure 1: Basic implementation of systolic array without pipelining (eyeriss architecture)

3.3 Line Buffers (Row Buffers)

Convolution requires three rows of image data at a time.
Therefore, we use **3 line buffers** implemented in BRAM:

- LineBuffer0 → Row y
- LineBuffer1 → Row y+1
- LineBuffer2 → Row y+2

These rows stream into the convolution array.

As new rows are fetched, older rows shift upward—this produces a continuous sliding window without halting computation.

4. **Dataflow Explanation** (Cycle-by-Cycle Behavior)

Step 1 — Input Streaming

Three consecutive rows of the input image are continuously streamed into the leftmost column of the PE array:

LB0 → PE(0,0)

LB1 → PE(1,0)

LB2 → PE(2,0)

Each cycle, pixels move right through systolic shifts.

Step 2 — Weight Preloading

Before convolution starts, weights W00–W22 are preloaded into the 3×3 stationary positions:

Row 0 PEs store W00, W01, W02

Row 1 PEs store W10, W11, W12

Row 2 PEs store W20, W21, W22

Weights remain fixed for the entire convolution.

Step 3 — MAC Computation inside the Array

At each PE:

$$PS_{out} = W_{ij} \cdot I(x, y) + PS_{in}$$

- Horizontal movement: pixels shift right
- Vertical accumulation: partial sums move down

After 3 vertical accumulations, PE(2, col) outputs the final convolution result.

Step 4 — Sliding Window Operation

Every cycle:

- New pixel enters from the left
- All pixels shift right
- Partial sums propagate downward

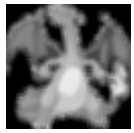
The window slides horizontally every cycle, then vertically after each row shift (after a newline interrupt).

5. Output Generation

Once vertical accumulation completes at the bottom PE row, the final convolved pixel is written to the **output image buffer**.

Output is generated **streamingly**, one pixel per cycle after pipeline fill.

5.1 Some convolution outputs using standard filters

Input image (64 x 64)		**Kernal**
Grayscale		-
Gaussian Blur		$\frac{1}{16}\begin{bmatrix}1 & 2 & 1\\ 2 & 4 & 2\\ 1 & 2 & 1\end{bmatrix}$
Sobel edge detection		$\begin{bmatrix}-1 & -1 & -1\\ 0 & 0 & 0\\ -1 & -1 & -1\end{bmatrix}$
Identity kernel		$\begin{bmatrix}0 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 0\end{bmatrix}$
High intensity		$\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$

6. Performance Characteristics

A conventional convolution implementation processes each output pixel independently, requiring nine multiply–accumulate (MAC) operations for a 3×3 kernel and repeating this computation 4096 times for a 64×64 image (excluding borders). This results in **36,864 MAC operations** executed sequentially or in small batches, depending on available hardware parallelism. Traditional architectures typically fetch all nine pixels of every 3×3 window from memory, perform nine multiplications, sum them, and then move to the next pixel. This method has two primary drawbacks: **high memory bandwidth demand** and **low computational reuse**, because each pixel is fetched multiple times by overlapping kernels. Memory access becomes the bottleneck, and throughput is limited to roughly **one output every 9 cycles** in a single-MAC design or **one output every 3–4 cycles** in a small parallel architecture.

In contrast, the systolic array architecture restructures the computation into a **streaming pipeline** containing **nine PEs arranged as three vertical 1×3 MAC chains**. This approach exploits both spatial and temporal data reuse: each input pixel participates in **three convolution columns** but is loaded only once into the line buffer. After the initial pipeline fill (~200 cycles), the design achieves **one complete 3×3 convolution output per clock cycle**, offering a theoretical throughput improvement of **9×–12×** over a traditional sequential MAC design.

Data movement efficiency is a major gain. Instead of loading nine pixels per output, the systolic array loads **one pixel per cycle from memory**, and the line buffers automatically provide the remaining eight pixels by shifting data internally. This reduces DRAM bandwidth by a factor of **8–9×**, enabling real-time or near–real-time processing even on low-power FPGA platforms. Additionally, partial sums flow forward through the systolic structure, allowing all MAC units to remain active every clock cycle, which yields a sustained **PE utilization close to 100%**—far higher than typical traditional designs where many cycles are spent waiting for memory.

Overall, for a 64×64 image, traditional convolution may require **36k–40k cycles**, while your systolic architecture can complete the same operation in approximately **4096 cycles + pipeline latency**, demonstrating an order-of-magnitude performance advantage in both computation and memory efficiency.

Metric	Traditional Convolutional Architecture	Systolic Convolutional Architecture
Image Size	64×64	64×64
Kernel Size	3×3	3×3
Total MAC Operations	36,864 (64×64×9)	Same: 36,864, but parallelized
MAC Units Active per Cycle	1–3 (typical)	9 (all PEs active every cycle after warm-up)
Memory Reads per Output Pixel	9 pixel reads	1 pixel read (8 reused by line buffers)
Memory Bandwidth Requirement	High (≈ 9× per-pixel requirement)	Very low (≈ 1/9 of traditional)
Data Reuse	Minimal; repeated fetching	Very high; reuse in line buffers + systolic flow
PE Utilization	20–40% typical	~100% after pipeline fill
Pipeline Latency (Initial)	None (non-pipelined)	~195–200 cycles to fill line buffers + PEs
Throughput (Outputs per Cycle)	1 output / 9 cycles (≈0.11 px/cycle)	1 output / 1 cycle (1 px/cycle)
Total Cycles for 64×64 Conv	~36,000–40,000 cycles	4096 cycles + ~200 latency ≈ 4300 cycles
Speed-up Factor	Baseline	8×–10× faster
Memory Bottleneck	Yes (dominant)	No (amortized by local reuse)
Computation Bottleneck	Yes (single MAC limits speed)	No (fully parallel 9-MAC systolic chain)
Scalability	Limited by MAC count and memory bandwidth	Highly scalable in both dimensions (more PEs → more outputs/cycle)
Power Efficiency	Moderate; many refetches waste energy	High; locality + reuse minimize DRAM energy
Architecture Type	Sequential / partially parallel	Pipelined, streaming, systolic dataflow

Table 1: Performance breakdown between normal convolution and systolic array architecture

7. Pipelining

The upgraded memory-loading system replaces block-based data delivery with a fully pipelined, continuous data-flow mechanism. The architecture uses **line buffers** and **sequential memory streaming** to generate a new 3×3 convolution window every cycle. This eliminates stalls inside the processing array and achieves much higher throughput than earlier row-by-row or block-based methods.
The system operates in two phases:

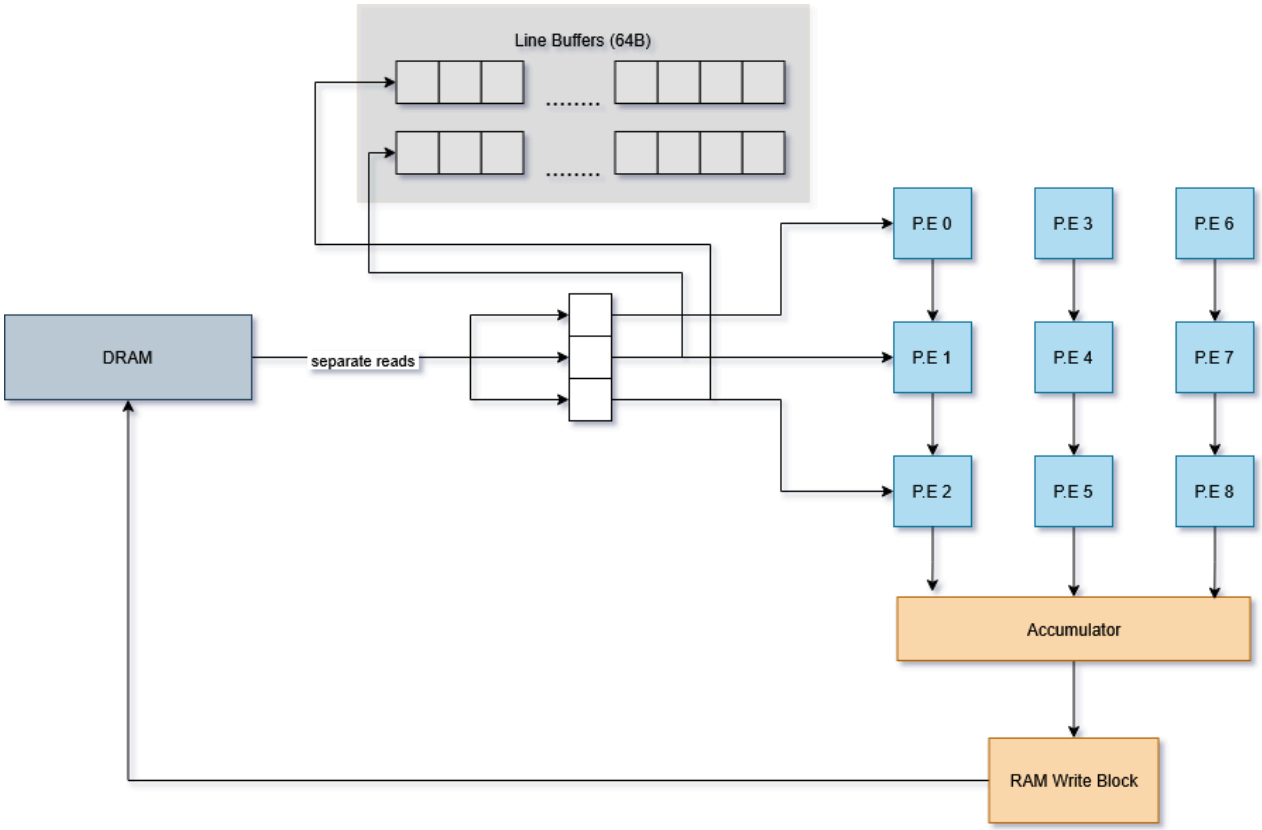


Figure 2: Pipelined systolic array architecture

(A) Initialization Phase

- Three rows of data are fetched sequentially.
- Line buffers store two previous rows while the third row streams in.
- After this warm-up, a complete 3×3 window becomes available.

(B) Steady-State Sliding Window Phase

- Every new pixel from memory slides the 3×3 window by one position.
- Eight of the nine pixels in the window come from the line buffers (reuse).
- Only one new pixel from memory is needed per cycle.

- Once primed, the system provides **one complete convolution window every clock cycle**.

The performance gain grows with image size because reuse increases with larger feature maps.

For a 32×32 input

- Old memory reads: ~2700 cycles
- New memory reads: ~900 cycles
- 3 x speedup

For a 128×128 input

- Old memory reads: ≈ 147,000
- New memory reads: ≈ 49,000
- 3 × less memory traffic, ~3× throughput gain

For a 224×224 input (ImageNet)

- Old memory reads: ≈ 450,000
- New memory reads: ≈ 150,000
- Saves ~300,000 RAM accesses

8. Summary

In this project, we designed and implemented a **systolic-array based convolution engine** for performing 3×3 convolution on images. Leveraging the principles of a **systolic array architecture**, in particular inspired by the dataflow used in Eyeriss, the design efficiently streams input image rows through a 3-row processing element (PE) grid. Kernel weights remain stationary in the PE rows, while input pixels and partial sums flow horizontally and vertically, respectively. This structure enables high data reuse for both input pixels and weights, reduces memory bandwidth demands, and exploits pipelined parallelism to produce one output pixel per cycle (after pipeline fill). We validated the design by convolving input images with a 3×3 kernel and obtained correct output images, demonstrating both functional correctness and efficacy of the hardware-oriented convolution method.

9. Conclusion

From the results and implementation, we conclude that our systolic-array convolution engine:

- Successfully implements 3×3 convolution in a hardware-friendly manner, achieving efficient data reuse and pipelined computation without the need for redundant memory accesses.

- Demonstrates that a row-stationary dataflow in a 3-row PE array (as per Eyeriss-style design) is effective for small-kernel convolution, combining simplicity, regular structure, and performance

benefits.

- Validates correctness by producing expected output images for given inputs, confirming that the architecture, line-buffers, data streaming logic, and MAC + accumulation operations work as intended.

However, like any practical design, the current implementation has scope for further improvement. In future work, the project can be extended/improved in the following ways:

- **Support for larger kernels or multiple channels**: Extend the design beyond a fixed 3×3 kernel support for arbitrary kernel sizes (e.g. 5×5, 7×7) or different kernel shapes, or multi-channel (e.g. RGB) input images.

- **Parallelism / array scaling**: Expand the PE array (both in number of rows and columns) to support larger images, higher throughput, or batch processing. This can include tiling, folding, or hierarchical systolic array organization for scalability.

- **Flexible dataflows / mixed stationary strategies**: Incorporate alternative dataflows (e.g. weight-stationary, output-stationary, or hybrid) to better suit different convolution parameters or workloads, thereby improving resource utilization or throughput.

- **Resource optimization and hardware synthesis (FPGA / ASIC)**: Translate the design into hardware description, synthesize and map onto FPGA or ASIC measure resource usage (LUTs, DSPs, memory), timing, and power / performance metrics.

- **Support for full convolutional layers / pipelines**: Expand from standalone convolution to complete convolutional layers — integrate with pooling, activation functions, multiple channels, and possibly multiple layers to approximate a lightweight CNN accelerator.

- **Performance evaluation and benchmarking**: Quantitatively analyze throughput, latency, memory bandwidth, resource utilization. Compare with naïve software convolution or other hardware implementations to demonstrate performance gains.

Overall, the project shows that a systolic-array convolution engine is a feasible and efficient approach for hardware-accelerated convolution. With further development along the directions above, it could form the basis for a more comprehensive and capable convolutional accelerator suitable for FPGA or ASIC deployment, and potentially extendable to full convolutional neural network use-cases.

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